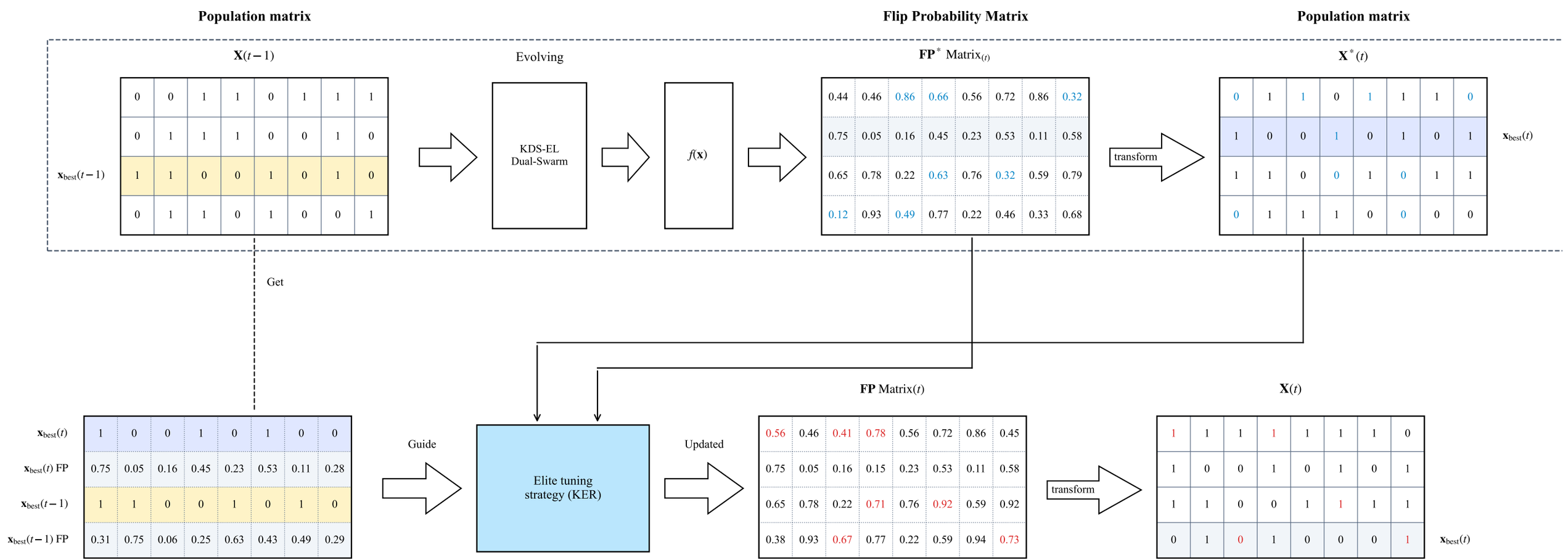
Population matrix
Flip Probability Matrix
Population matrix
X(t − 1)
Evolving
FP* Matrix(t)
X*(t)
0 0 1 1 0 1 1 1
0 1 1 1 0 0 1 0
1 1 0 0 1 0 1 0
0 1 1 0 1 0 0 1
x_best(t − 1)
KDS-EL
Dual-Swarm
f(x)
0.44 0.46 0.86 0.66 0.56 0.72 0.86 0.32
0.75 0.05 0.16 0.45 0.23 0.53 0.11 0.58
0.65 0.78 0.22 0.63 0.76 0.32 0.59 0.79
0.12 0.93 0.49 0.77 0.22 0.46 0.33 0.68
transform
0 1 1 0 1 1 1 0
1 0 0 1 0 1 0 1
1 1 0 0 1 0 1 1
0 1 1 1 0 0 0 0
x_best(t)
Get
x_best(t) 1 0 0 1 0 1 0 0
x_best(t) FP 0.75 0.05 0.16 0.45 0.23 0.53 0.11 0.28
x_best(t − 1) 1 1 0 0 1 0 1 0
x_best(t − 1) FP 0.31 0.75 0.06 0.25 0.63 0.43 0.49 0.29
Guide
Elite tuning
strategy (KER)
Updated
FP Matrix(t)
X(t)
0.56 0.46 0.41 0.78 0.56 0.72 0.86 0.45
0.75 0.05 0.16 0.15 0.23 0.53 0.11 0.58
0.65 0.78 0.22 0.71 0.76 0.92 0.59 0.92
0.38 0.93 0.67 0.77 0.22 0.59 0.94 0.73
transform
1 1 1 1 1 1 1 0
1 0 0 1 0 1 0 1
1 1 0 0 1 1 1 1
0 1 0 1 0 0 0 1
x_best(t)